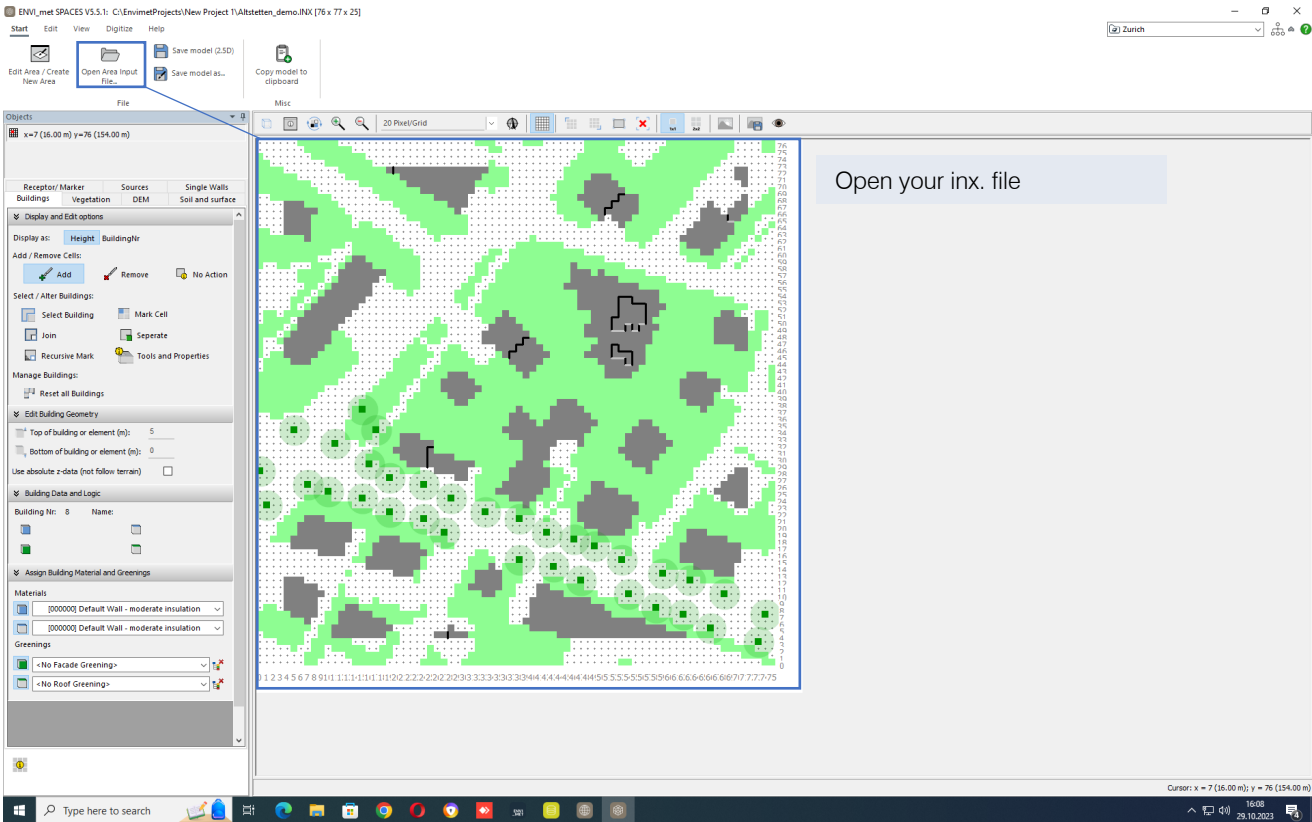
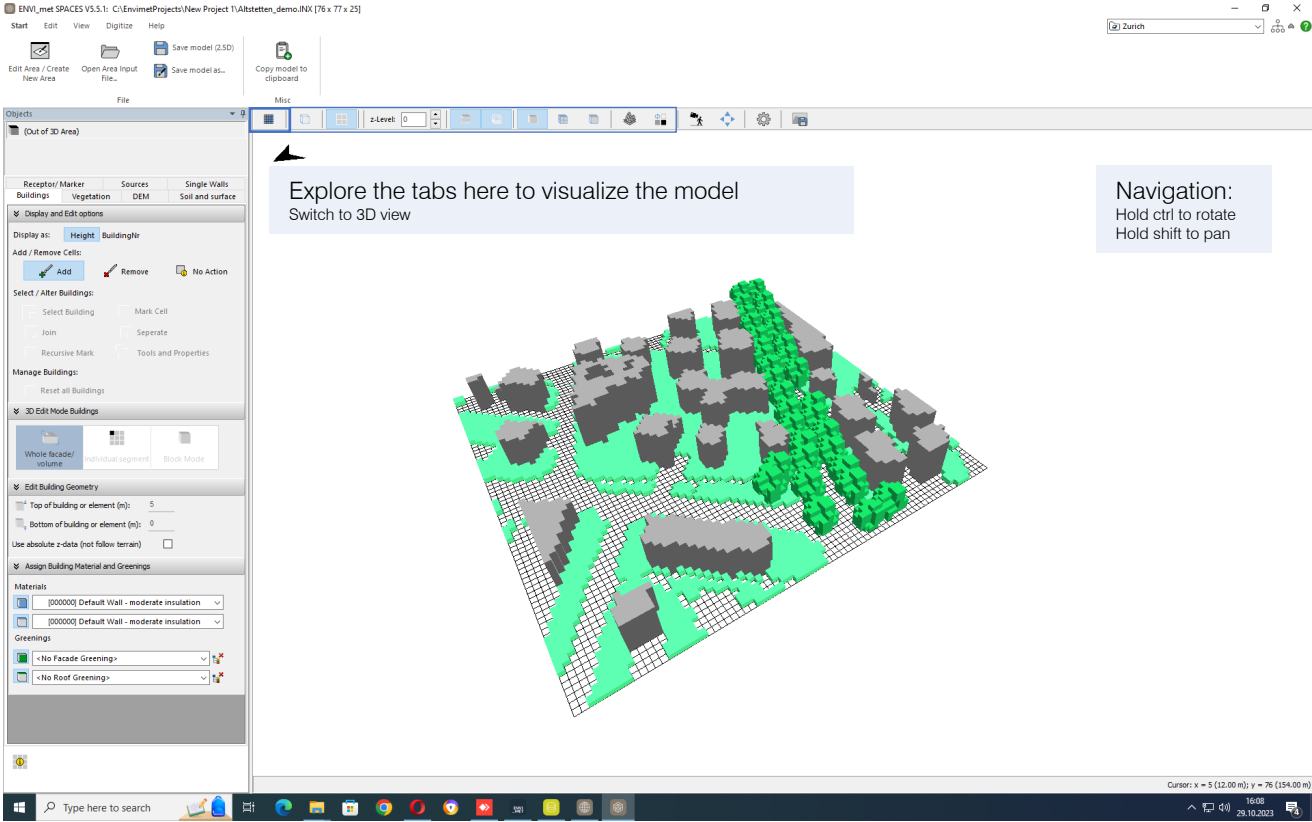


Digitizing with Spaces

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Digitizing with Spaces

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ENVI-met Model Analyser

Model Geometry Overview

Model Geometry

Grid dimensions: 76 x 77 x 25 Grids
dx=2.00 m dy=2.00 m
Base dx=2.00 m
Core XY domain size: 152.00 m x 154.00 m

Objects and Terrain:

Highest building in domain: 20 m
Highest DEM point in domain: 0 m [ref]
Highest point building + DEM: 20 m

Automatic Grid Generation (Concept Design)

Horizontal

0 Nesting grids at each border

Resulting dx-dy spacing:

Border complete model
Border core model
dx=2.00 m from here

Vertical

Grid generation: Equidistant grid (lowest cell splitted into 5 sub-cells)
Base dx= 2.00 m. Sub-cell dx=0.40 m
Resulting dx spacing and height cell center:

Top of 3D model
z=10 dx=2.00 m, also z= 49.00 m
z=24 dx=2.00 m, also z= 47.00 m
z=32 dx=2.00 m, also z= 45.00 m
z=32 dx=2.00 m, also z= 43.00 m
z=32 dx=2.00 m, also z= 41.00 m
z=32 dx=2.00 m, also z= 39.00 m
z=18 dx=2.00 m, also z= 37.00 m
z=18 dx=2.00 m, also z= 35.00 m

(also, z refers to grid box center)

Geometry Check

Check distance buildings to model border

Min distance betw. buildings and model border: 4 Grids
Extra space added by nesting area: 8.00 m
Resulting total minimal space in the model: 8.00 m
Only very little space between buildings and model border!

Vertical Extension

Height of 3D model top: 50.00 m
Highest point building + DEM: 20 m
Difference model top to highest point: 30.00 m
Sufficient space between Building/DEM top and model border.

Adjust Geometry and Spacing

Horizontal

Add empty grid cells at the core border

Apply

Vertical

Start telescoping after height [m]: 0
Telescoping factor [R]: 0
Try Apply to model

Model Analyser

The Model Analyser summarizes the geometry of your model and performs basic checks of the aspect ratios between the model itself and the model borders.

Automatic Grid Generation

In Concept Design Mode, the model grids in the horizontal Nesting Grid area (if used) and the layout of the vertical grids are generated based on parameters given for the .INX file.

Geometry check

As a rule of thumb, the horizontal distance between the border and the buildings should be either zero or at least the size of the closest building height. For the vertical spacing, sufficient space must be between the highest point of the model and the model border. Best results come, if the distance equals the height of the highest element in the model.

Adjust Geometry

This allows a quick fixing of spacing issues by adding additional grid cells or changing the settings for vertical grid generation. Changes in the vertical grid only work in Concept Design mode as in Detailed Design the model grid layout is fixed.

Run the model inspector Geometry check: Make sure that your buildings and the model border has enough space in between. If not, go back to Monde and select a larger sub area.

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Adjust Geometry

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Go to "soil and surface" tab and apply a soil material to all grids.

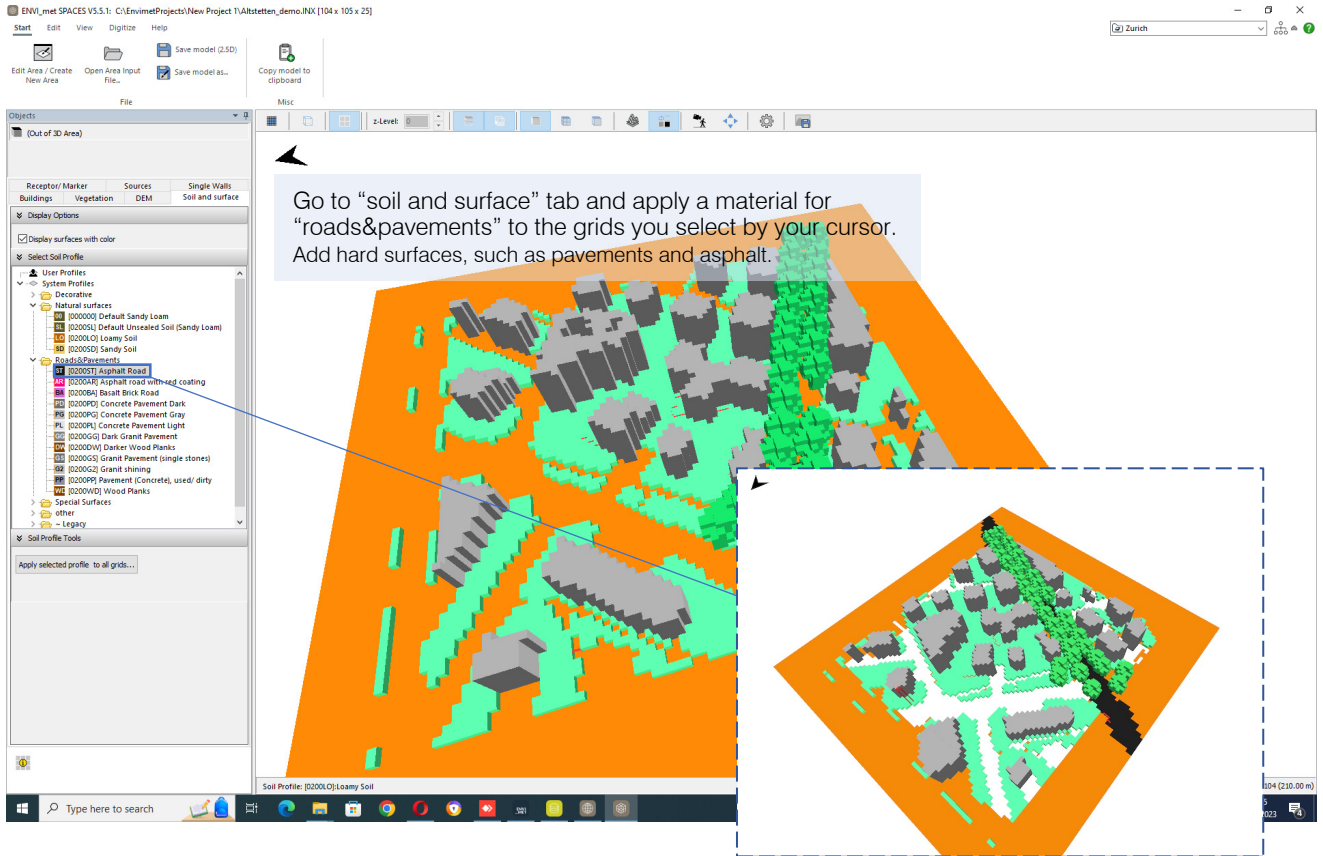
Soil Profile Tools

Apply selected profile to all grids...

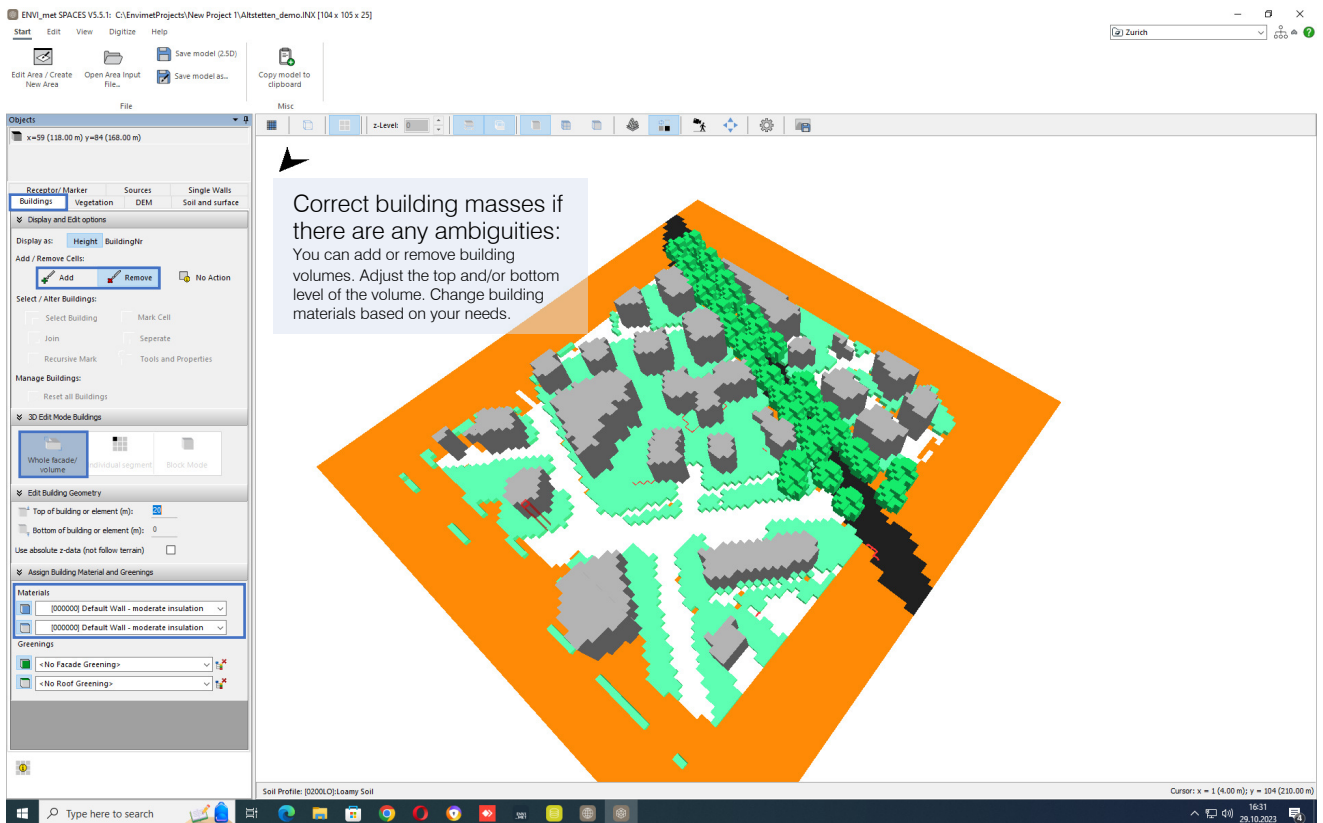
Soil Profile: 000002 Default Sandy Loam

Digitizing with Spaces

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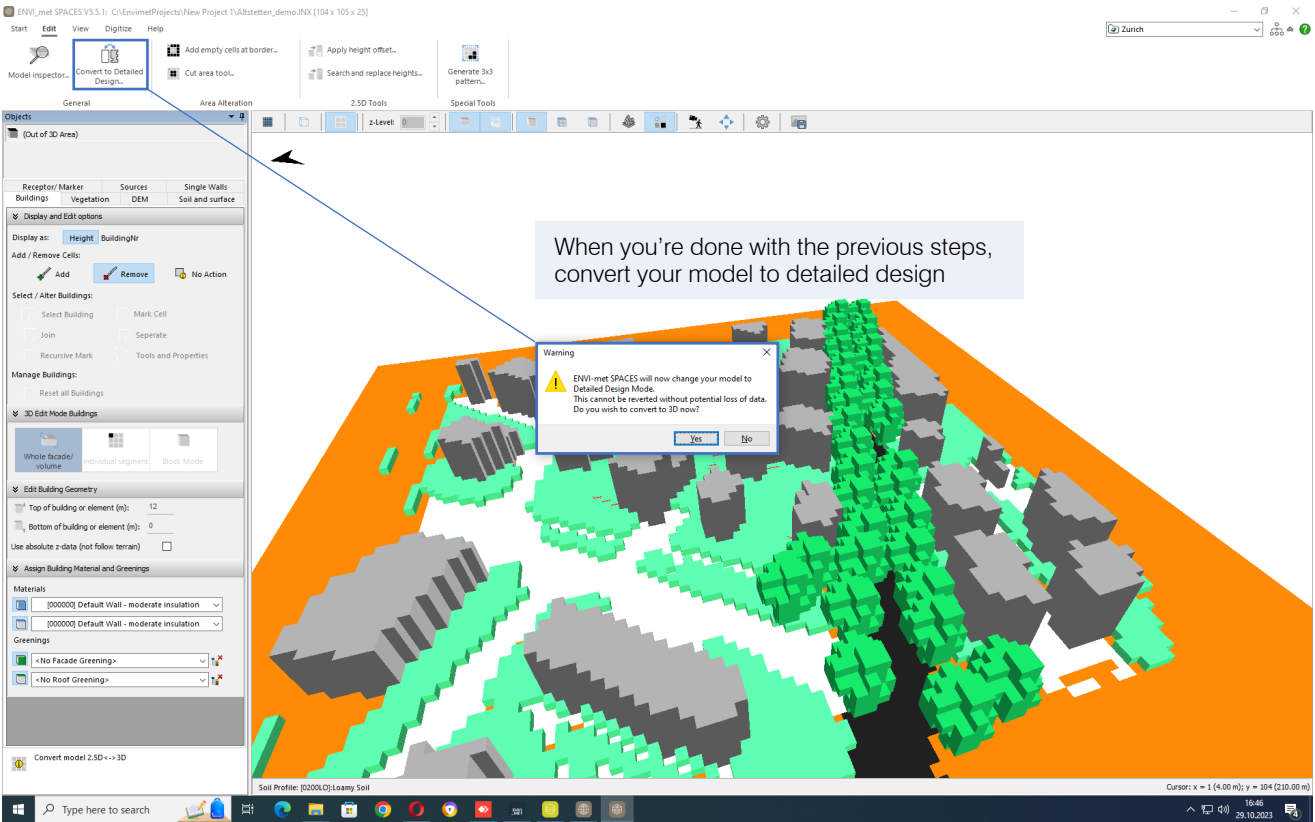


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Digitizing with Spaces

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